

Unlocking Flexibility Potential through DVFS-controlled Data Center Clusters in Unit Commitment

Abstract—Data center clusters (DCCs) have become recognized as significant energy consumers in electrical power system operations. Workload balancing for DCCs, implemented by dynamic voltage and frequency scaling (DVFS) control schemes, allows for providing flexibility potential capability and contributes to maintaining the security of scheduling results through Unit Commitment (UC). To achieve this, the flexibility potential of DCCs under different workload conditions is analyzed. Building on this analysis, a new set of physical constraints representing workload balancing across multiple DVFS-controlled DCCs is proposed, and related linearized constraints is extended through McCormick Envelopes method. Furthermore, these intractable constraints are incorporated into UC, and an enhanced UC (EUC) model, which considers the flexibility potential offered by DVFS-controlled DCCs, is developed. Several experiments are conducted on the IEEE 6 bus and 118 bus test systems to assess the validity and scalability of the proposed models.

Index Terms—Data center clusters, Flexibility Capability, Enhanced Unit Commitment, McCormick Envelopes

I. INTRODUCTION

AS the increase in various artificial intelligence technologies and demand energy consumption, the operation of electrical power systems are becoming diverse and uncontrollable, and they are facing challenges in provide sufficient flexible adjustment resources against emergence events [1], [2]. In such circumstances, additional flexible mechanisms such as demand response, virtual Power plant, and energy storage systems, etc., needs to be prioritized. Nevertheless, as a significant energy consumer, DCCs have not been fully utilized in the flexibility potential of power system operations. To this end, it is essential to investigate the flexible adjustment characteristic of DCCs, and to explore the potential of DCCs in providing flexible adjustment resources for the economic and reliable operation of electrical power systems.

Recent literature reported several studies on the DCCs flexibility modelling and its application in electrical power systems operation. Specially, Ref. [3] carried out a Quality-of-service-based cost function to describe the workload shifting and server utilisation of DCCs, and extended a distributed dispatch framework with considering DCCs flexibility. Ref. [1] designed a mix-integer programming model to achieve the optimal demand management of DCCs. Ref. [2] analysed the progressive load characteristics in DCCs, and proposed two optimization models to estimate the optimal capability planning of storage systems in DCCs and design parameters optimization. Ref. [4] explored the statistical feasibility of DCCs and hydrogen resources in a microgrid context, and carried out a data-driven statistical-feasibility-based robust rolling optimization framework to analysis the optimal operation of DCCs and hydrogen storage systems under complex uncertain factors. Ref. [5] further analysed the potential capability of DCCs for regulating computational workloads while reducing carbon

emissions, and constructed a emission-aware coordinated dispatch method of multi-regional microgrid with DCCs. The aforementioned studies primarily emphasise the flexibility capabilities of DCCs within electrical power system operations. However, they often overlook the intricate physical characteristics of DCCs, particularly in relation to workload shifting dynamics and the implementation of advanced high-evolution control strategies.

The workloads in DCCs can be broadly classified into three categories, [1], *i) compute-intensive workloads*, which include high-performance computing (HPC), machine learning model training, and commercial application services. *ii) storage-intensive workloads*, which are employed for tasks such as regular file backups, archival systems, and relational database applications. *iii) network-intensive workloads*, encompassing activities like website hosting, content delivery networks, and network traffic management. Notably, compute-intensive and storage-intensive workloads dominate in DCCs, collectively accounting for approximately 70% of the total workload. This dominance is driven by the growing demand for HPC and data-driven techniques. In contrast, network-intensive workloads are typically less predictable and highly time-sensitive, requiring immediate processing to ensure seamless operation. Since the former two workloads are more predictable and flexible, the peak workload requirements of the former two workloads can be effectively shifted or local workload delay responded through DVFS, [1], [2], [6], to achieve spatio-temporal workload balancing and total energy management.

In addition to the energy management process from DCCs using DVFS control schemes, additional physical constraints concerning task-size bandwidth and time-tolerant workload types must be taken into account, as they directly influence the feasibility of workload shifting between multiple DCC servers and their operational logic. For instance, excessively shifting time-sensitive workloads, particularly network-intensive ones, between DCCs may lead to potential risks such as data congestion and latency due to limited transmission bandwidth, thereby compromising the ability to meet actual real-time demands. To reflect this issue, Ref. [7] investigated a queuing model implemented in cloud-based DCCs using DVFS load-dependent strategies, and highlighted its significant on energy efficiency and workload management. Ref. [8] analysed the response characteristics of various time-tolerant tasks on workload completion time and total energy consumption, and proposed a dual-timescale collaborative optimization model for multi-DCC systems. Ref. [9] explored the impact of batch workload scheduling in DCCs on the energy consumption, and developed a day-ahead emission-aware stochastic optimization model for microgrid energy planning and generation management. Ref. [10] extended a data-driven distributionally robust optimization framework for low-carbon DCC systems,

addressing multi-tasking responses and uncertainties in renewable energy integration. Ref. [6] proposed a workload-aware robust DVFS technique, termed as GearDVFS, to address concurrency challenges in mobile-driven workloads. Ref. [11] provided a comprehensive survey of efficient task scheduling and resource allocation methodologies in multi-DCC networks. Ref. [12] proposed a distributed shortest remaining time first (SRTF) scheduling model for DCCs, designed to minimize flow completion time while ensuring flow-size availability. Ref. [13] constructed a machine-learning-based scheduling flow detection and packet-switched optical network (PSON) architecture with centralized control for intra-DCC connectivity and bath workload allocation. These studies comprehensively address multi-task assignments and flow-size bandwidth limitations across different DCCs, further developing various task allocation models and analysing the energy consumption of received computing tasks. However, the potential interaction mechanisms between workload processes, particularly those involving multi-task shifting across multiple DCCs and their role in meeting electrical flexibility requirements, remain underexplored.

Driven by this motivation, this paper analyses the flexibility offered by DVFS-controlled DCCs and integrates this capability into UC studies. A key aspect of this work is the physical modeling of the interaction mechanism between DCC energy consumption and workload balancing strategies managed by DVFS. Building upon this foundation, additional restrictions such as bandwidth limitations and time-delay requirements for various workload types are reshaped into corresponding physical constraints and linearized reformulated. In contrast to previous studies that focused on workload shifting strategies or energy consumption reduction, this research provides a comprehensive analysis of the coordinated optimization between power system operations and the DCC workload execution process. Moreover, the multi-timescale interactions between active response characteristic of DCCs through workload shifting and grid-side demand fluctuation is modeled in detail, considering both grid-side requirements and specific implementation details of workload balancing processes.

The main contributions of this paper are summarized below: *i)* A DVFS-controlled physical constraints is developed to characteristic the flexibility attributes of DCCs workloads. Furthermore, the proposed constraints are linearized using McCormick Envelopes to enhance computational tractability in UC studies. *ii)* A EUC model is formulated to capture the dynamic flexible response of DCCs. The model incorporates a new coordination mechanism to address multi-timescale interactions between workload shifting in DCCs and grid-side demand fluctuations.

The remainder of this paper is organized as follows. Section II introduces the proposed scheduling framework, which integrates the flexibility potential of DVFS-controlled DCCs. Section III develops a new set of physical constraints to model DCC workload shifting and local energy consumption requirements. Additionally, the McCormick Envelope method is employed to linearized the nonlinear constraints, enabling the estimation of DCC flexibility potential. Section IV incorporates the flexibility potential of DVFS-controlled DCCs into UC

and formulates the proposed EUC model in detail. Section V presents numerical experiments conducted on benchmark test systems to validate the effectiveness and scalability of the proposed model. Finally, Section VI summaries the key findings and insights derived from the case studies.

II. THE PROPOSED SCHEDULING FRAMEWORK

This section presents a coordinated scheduling framework to integrate the flexible response capabilities of DCCs into electrical power system operations. Recognising the difference in scheduling timescales (hourly for grid-side operations versus seconds for DCCs workload management), the framework is structured into two interconnected parts. PART-1 encompasses the grid-side scheduling model, while PART-2 represents the DCCs workload management model.

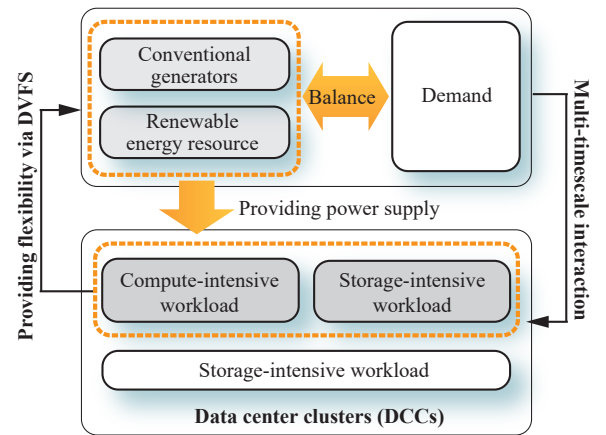


Fig. 1. The proposed scheduling framework in this work

III. REFORMULATED CONSTRAINTS FOR DVFS-CONTROLLED DCCs

The power consumption of DCCs is primarily influenced by two components: static energy and dynamic energy. The static energy, often referred to as idle power, arises mainly from the leakage current in the CPU core circuits. In contrast, the dynamic energy is predominantly determined by the workload ratio λ_{dcc} and the individual service level μ_{dcc} . The total power consumption of DCCs can be expressed as follows, [6],

$$p_{dcc,i} = p_{dcc,i}^{idle} + p_{dcc,i}^{dyn}, \forall i \in \mathcal{I} \quad (1a)$$

$$p_{dcc,i}^{idle} = v_{dcc,i} i_{leak}, \forall i \in \mathcal{I} \quad (1b)$$

$$p_{dcc,i}^{dyn} = (C_{dcc,i} f_{dcc,i} v_{dcc,i}^2) \lambda_{dcc,i} \mu_{dcc,i}^{-1}, \forall i \in \mathcal{I} \quad (1c)$$

where i and $\mathcal{I}$ represent the index and set of DCCs. $p_{dcc,i}$ denotes the total power consumption of individual DCC i . $p_{dcc,i}^{idle}$ and $p_{dcc,i}^{dyn}$ represent the static (idle) and dynamic power consumption of DCC i , respectively. $C_{dcc,i}$ defines the constant representing the effective capacitance of DCC i . $f_{dcc,i}$ and $v_{dcc,i}$ denote the operational frequency and voltage of DCC i , respectively.

Based on the above physical working principles, the set of physical security constraints representing DCCs workflows can be extended,

$$p_{dcc,i} \geq p_{dcc,i}^{idle} + p_{dcc,i}^{dyn}, \forall i \in \mathcal{I} \quad (2a)$$

$$p_{dcc,i}^{idle} \geq \lfloor v_{dcc,i} \rfloor i_{leak}, p_{dcc,i}^{idle} \leq \lceil v_{dcc,i} \rceil i_{leak}, \forall i \in \mathcal{I} \quad (2b)$$

$$p_{dcc,i}^{dyn} \geq \lfloor \phi_{dcc,i}^{dvfs} \rfloor \lambda_{dcc,i} \mu_{dcc,i}^{-1}, p_{dcc,i}^{dyn} \leq \lceil \phi_{dcc,i}^{dvfs} \rceil \lambda_{dcc,i} \mu_{dcc,i}^{-1}, \quad (2c)$$

$$\phi_{dcc,i}^{dvfs} = (C_{dcc,i} f_{dcc,i} v_{dcc,i}^2), \forall i \in \mathcal{I}$$

where $\lceil \circ \rceil$ and $\lfloor \circ \rfloor$ represent pre-determined upper and lower bounds for the decision variable $\circ$. The term $\phi_{dcc,i}^{dvfs}$ represents the power regulation process for the DCC i through DVFS technique, while $\lambda_{dcc,i}$ describes the workload balancing process for the DCC i . The feasibility of workload balancing for compute-intensive and storage-intensive workloads hinges on users' latency tolerance for batch workloads and the capacity to shift workloads among DCCs. When users accept tolerate higher latency, the execution of workloads can be dynamically adjusted across multi DCCs. By optimizing these workload execution patterns, the overall energy consumption of DCCs can be optimised through DVFS and sequentially reduced, thereby enhancing their demand response capabilities for power system operations.

To reflect it, additional concise constraints concerning workload balancing processes among multi DCCs are constructed,

$$\Delta p_{dcc,i}^{bal} = \phi_{dcc,i}^{dvfs} (\lambda_{dcc,i}^{ini} + \Delta \lambda_{dcc,i}) \mu_{dcc,i}^{-1}, \forall i \in \mathcal{I} \quad (3a)$$

$$p_{dcc,i} \geq p_{dcc,i}^{idle} + p_{dcc,i}^{dyn} + \Delta p_{dcc,i}^{bal}, \forall i \in \mathcal{I} \quad (3b)$$

$$\sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}} \Delta \lambda_{dcc,i} = 0, \forall i \in \mathcal{I}, \forall t \in \mathcal{T} \quad (3c)$$

$$\lfloor f_{dcc,i} \rfloor \leq f_{dcc,i} \leq \lceil f_{dcc,i} \rceil, \forall i \in \mathcal{I} \quad (3d)$$

$$\lfloor v_{dcc,i} \rfloor \leq v_{dcc,i} \leq \lceil v_{dcc,i} \rceil, \forall i \in \mathcal{I} \quad (3e)$$

where $\Delta p_{dcc,i}^{bal}$ represents the change in the power consumption for DCC i due to workload balancing, $\lambda_{dcc,i}^{ini}$ represents the initial workload, and $\Delta \lambda_{dcc,i}$ represents the adjusted workload. By incorporating the workload balancing dynamics defined in (3a)-(3c) into (2b) and (2c), the form of DVFS-controlled DCCs constraints can be established. Despite these constraints, these constraints remains elusive due to the presence of product terms involving continuous variables $\phi_{dcc,i}^{dvfs}$ and $\Delta \lambda_{dcc,i}$ in (3a), therefore, the McCormick Envelope method is further extended and then integrated into UC studies. On this basis, an union form of EUC considering DVFS-controlled DCCs is listed below,

$$\text{objective function : minimize operation cost} \quad (4a)$$

$$\text{security constraints : } \begin{cases} \text{system-wide power balance,} \\ \text{units output bindings,} \\ \text{and network power flow, etc.} \end{cases} \quad (4b)$$

$$\text{DCCs constraints : (2b), (2c), and (3a) - (3e)} \quad (4c)$$

REFERENCES

- [1] J. Li, Z. Li, K. Ren, and X. Liu, "Towards optimal electric demand management for internet data centers," *IEEE Transactions on Smart Grid*, vol. 3, no. 1, pp. 183–192, 2012.
- [2] A. Wierman, Z. Liu, I. Liu, and H. Mohsenian-Rad, "Opportunities and challenges for data center demand response," in *International Green Computing Conference*, 2014, pp. 1–10.

- [3] A. Tsiligkaridis, P. Andrianesis, A. K. Coskun, M. C. Caramanis, and I. C. Paschalidis, "Distributed economic dispatch in power networks incorporating data center flexibility," *IEEE Transactions on Sustainable Computing*, pp. 1–15, 2025.
- [4] J. Duan, Y. Li, Y. Li, P. Yang, and Z. Zeng, "Statistical feasibility robust optimization with polyhedron uncertainty set for hydrogen-data center microgrid operations," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 9884–9902, 2025.
- [5] X. Wen, J. Zhu, and H. Dong, "Carbon emission-aware coordinated dispatch of multi-regional microgrids with data centers," *IET Conference Proceedings*, vol. 2024, pp. 1014–1019, 2025.
- [6] C. Lin, K. Wang, Z. Li, and Y. Pu, "A workload-aware dvfs robust to concurrent tasks for mobile devices," in *Proceedings of the 29th Annual International Conference on Mobile Computing and Networking*, 2023, pp. 1–16.
- [7] P. J. Kuehn and M. Mashaly, "DVFS-power management and performance engineering of data center server clusters," in *2019 15th Annual Conference on Wireless On-demand Network Systems and Services (WONS)*. Wengen, Switzerland: IEEE, 2019, pp. 91–98.
- [8] J. Han, Y. Yan, Y. Wang, K. Han, Y. Han, and J. Lin, "Dual-time scale collaborative optimization of data center energy system: Considering multi-task response mechanism and hybrid hydrogen-battery energy storage," *Journal of Energy Storage*, vol. 119, p. 116244, May 2025.
- [9] Z. Ding, L. Xie, Y. Lu, P. Wang, and S. Xia, "Emission-aware stochastic resource planning scheme for data center microgrid considering batch workload scheduling and risk management," *IEEE Transactions on Industry Applications*, vol. 54, no. 6, pp. 5599–5608, Nov. 2018.
- [10] J. Han, K. Han, T. Han, Y. Wang, Y. Han, and J. Lin, "Data-driven distributionally robust optimization of low-carbon data center energy systems considering multi-task response and renewable energy uncertainty," vol. 102, p. 111937.
- [11] K. Wang, Q. Zhou, S. Guo, and J. Luo, "Cluster Frameworks for Efficient Scheduling and Resource Allocation in Data Center Networks: A Survey," vol. 20, pp. 3560–3580.
- [12] C. Gao, V. C. Lee, and K. Li, "D-srtf: Distributed shortest remaining time first scheduling for data center networks," *IEEE Transactions on Cloud Computing*, vol. 9, no. 2, pp. 562–575, 2021.
- [13] L. Wang, X. Wang, M. Tornatore, K. Kim, S. M. Kim, D. Kim, K. Han, and B. Mukherjee, "Scheduling with Machine-Learning-Based Flow Detection for Packet-Switched Optical Data Center Networks," *JOCN*, vol. 10, pp. 365–375, 2018.